
Dependence Testing

Allen and Kennedy, Chapter 3 thru Section 3.3.2

Main Theme

- Determining whether dependencies exist between two subscripted references to the same array in a loop nest
- Several tests to detect these dependencies

Exercise

```
for I = 1 to 9 do {  
  S1: A[3*I] = ...  
  S2: ... = A[2*I]  
}
```

- Does the above loop has any dependency?

```
for I = 1 to 9 do {  
  for J = 1 to 9 do {  
    S1:      A[3*I, 2*J] = ...  
    S2:      ... = A[2*I, 3*J]  
  }  
}
```

The General Problem

```
DO i1 = L1, U1
  DO i2 = L2, U2
    ...
    DO in = Ln, Un
      S1    A(f1(i1, ..., in), ..., fm(i1, ..., in)) = ...
      S2    ... = A(g1(i1, ..., in), ..., gm(i1, ..., in))
    ENDDO
  ...
ENDDO
ENDDO
```

$$f_i(\alpha) = g_i(\beta) \text{ for all } i, 1 \leq i \leq m$$

Basics: Conservative Testing

- Consider only linear subscript expressions
- Finding integer solutions to system of linear Diophantine Equations is NP-Complete
- Most common approximation is **Conservative Testing**
See if you can assert
"No dependence exists between two subscripted references of the same array"
- Never incorrect, may be less than optimal

Basics: Indices and Subscripts

Index: Index variable for some loop surrounding a pair of references

Subscript: A PAIR of subscript positions in a pair of array references

For Example:

$A(I, j) = A(I, k) + C$

$\langle I, I \rangle$ is the first subscript

$\langle j, k \rangle$ is the second subscript

Basics: Complexity

A subscript is said to be

- ZIV if it contains no index
- SIV if it contains only one index
- MIV if it contains more than one index

For Example:

$$A(5, I+1, j) = A(1, I, k) + C$$

First subscript is ZIV

Second subscript is SIV

Third subscript is MIV

Basics: Separability

- A subscript is separable if its indices do not occur in other subscripts
- If two different subscripts contain the same index they are coupled

For Example:

$$A(I+1, j) = A(k, j) + C$$

Both subscripts are separable

$$A(I, j, j) = A(I, j, k) + C$$

Second and third subscripts are coupled

Basics: Coupled Subscript Groups

- Why are they important?

Coupling can cause imprecision in dependence testing

```
DO I = 1, 100
S1      A(I+1,I) = B(I) + C
S2      D(I) = A(I,I) * E
ENDDO
```

Dependence Testing: Overview

- Partition subscripts of a pair of array references into separable and coupled groups

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- Partition subscripts of a pair of array references into separable and coupled groups
- Classify each subscript as ZIV, SIV or MIV
- For each separable subscript apply single subscript test.

Classify as ZIV/SIV/MIV

- Easy step
- Just count the number of different indices in a subscript

Applying Single Subscript Tests

- ZIV Test
- SIV Test
 - Strong SIV Test
 - Weak SIV Test
 - Weak-zero SIV
 - Weak Crossing SIV
- SIV Tests in Complex Iteration Spaces

ZIV Test

```
DO j = 1, 100  
S      A(e1) = A(e2) + B(j)  
ENDDO
```

e1,e2 are constants or loop invariant symbols
If $(e1 - e2) \neq 0$ No Dependence exists

Strong SIV Test

- Strong SIV subscripts are of the form

$$\langle ai + c_1, ai + c_2 \rangle$$

- For example the following are strong SIV subscripts

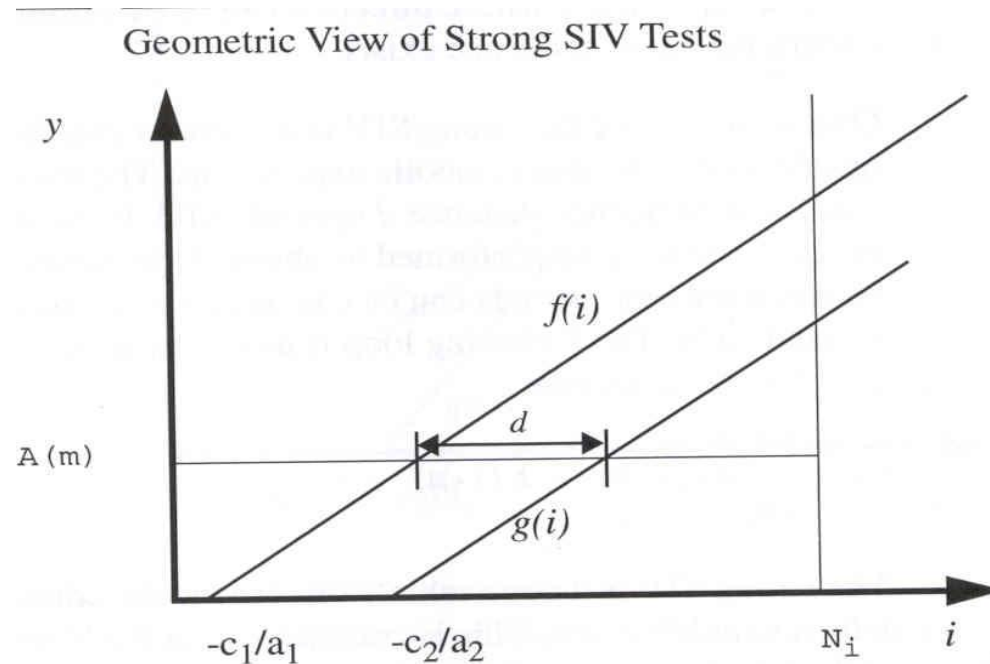
$$\langle i + 1, i \rangle$$

$$\langle 4l + 2, 4l + 4 \rangle$$

Strong SIV Test Example

```
DO k = 1, 100
  DO j = 1, 100
S1    A(j+1,k) = ...
S2          ... = A(j,k) + 32
  ENDDO
ENDDO
```

Strong SIV Test



$$d = i' - i = \frac{c_1 - c_2}{a}$$

Dependence exists if $|d| \leq U - L$

U and L are the loop upper and lower bounds

Strong SIV Test Example

```
DO j = 1, 100
S1  A(j+1) = ...
S2      ... = A(j+99) + 32
ENDDO
```

- Does the above loop has any dependency as per SIV test?

Weak SIV Tests

- Weak SIV subscripts are of the form

$$\langle a_1 i + c_1, a_2 i + c_2 \rangle$$

- For example the following are weak SIV subscripts

$$\langle i + 1, 5 \rangle$$

$$\langle 2i + 1, i + 5 \rangle$$

$$\langle 2i + 1, -2i \rangle$$

Weak-zero SIV Test

- Special case of Weak SIV where one of the coefficients of the index is zero
- The test consists merely of checking whether the solution is an integer and is within loop bounds

$$i = \frac{c_2 - c_1}{a_1}$$

Weak-zero SIV Test Example

```
      DO j = 1, 100  
S1      A(2*j-301) = ...  
S2      ... = A(55) + 32  
      ENDDO
```

- Does the above loop has any dependency as per Weak SIV test?

Weak-zero SIV & Loop Peeling

```
      DO i = 1, N  
S1      Y(i, N) = Y(1, N) + Y(N, N)  
      ENDDO
```

- Can we extract parallelism from the loop?

Weak-zero SIV & Loop Peeling

```
      DO i = 1, N
S1      Y(i, N) = Y(1, N) + Y(N, N)
      ENDDO
```

Can be loop peeled to...

```
      Y(1, N) = Y(1, N) + Y(N, N)
      DO i = 2, N-1
S1      Y(i, N) = Y(1, N) + Y(N, N)
      ENDDO

      Y(N, N) = Y(1, N) + Y(N, N)
```

Questions?